DEVIATHON 2.0

National Level Hackathon | GLA University, Mathura

AEGIS SAATHI

Problem Statement Title: AI-Powered Early Warning System for Seniors Living at Home

PS ID : DEV-104 | **Theme / Track:** TECH FOR IMPACT

Team Name: BROKECODERS

Team Member Names:

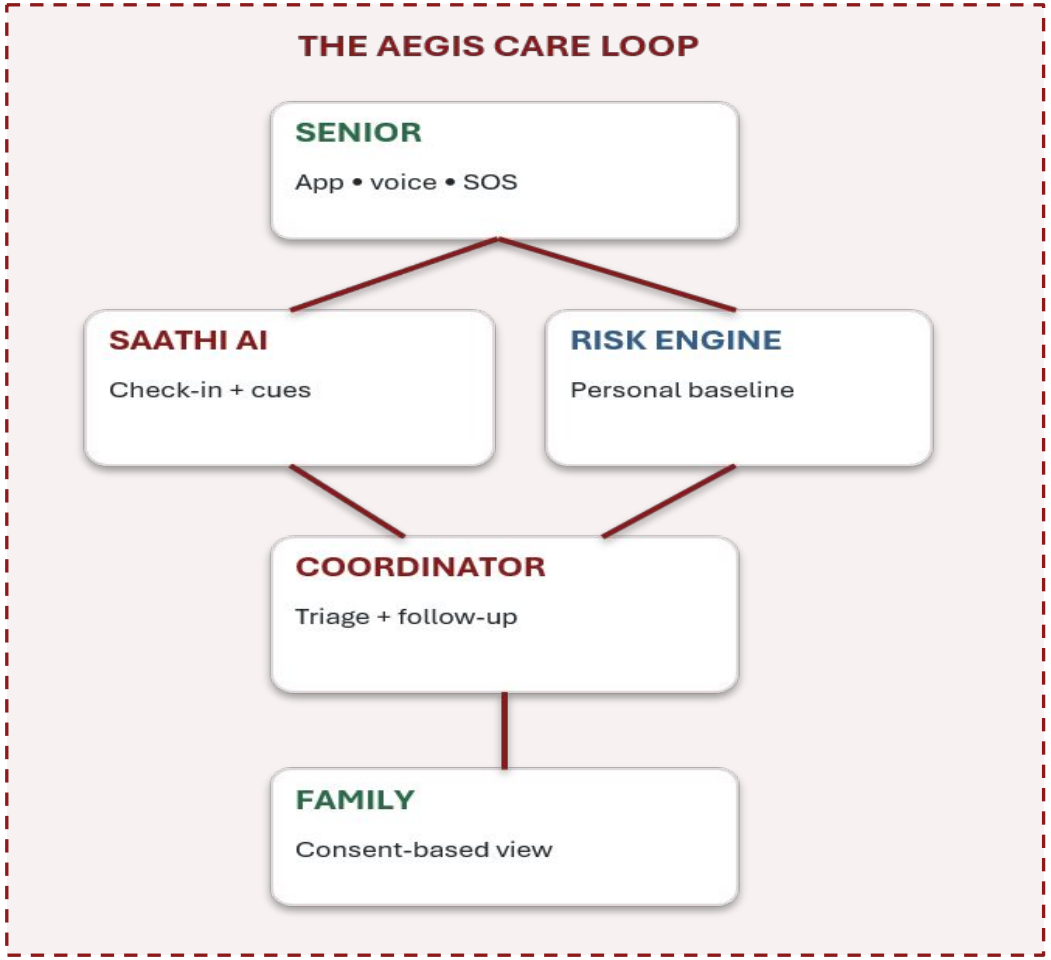
- Ark Agrawal
- Nandini Agarwal
- Adhish Gupta
- Navish Bharti

Github LINK - https://github.com/arkagrawall/Aegis-Care

AEGIS SAATHI

Proposed Solution (Describe your Idea/Solution/Prototype)

- **Closed-Loop Care System:** Seamless loop connecting senior check-ins → multi-tier risk engine → coordinator triage → family app notification.
- **Senior App & Saathi Voice:** Large tactile controls, Hindi/English toggle, daily vitals, and conversational voice check-in tracking pain, sleep & appetite.
- **Emergency Fall & SOS Protocol:** Impact + stillness detection launches a 30s audio safety check; failure to respond immediately escalates to coordinator.
- **Care Coordinator Triage:** Prioritized daily queue ranked by risk score with explainable clinical cues and one-click recommended response actions.
- **Family Reassurance Portal:** Consent-driven daily status streaks, step tracking, and peace of mind without overwhelming medical terminology.
- **Clinically Grounded Detection:** Catches subtle gradual decline patterns up to 9 days early while minimizing operational alert fatigue.



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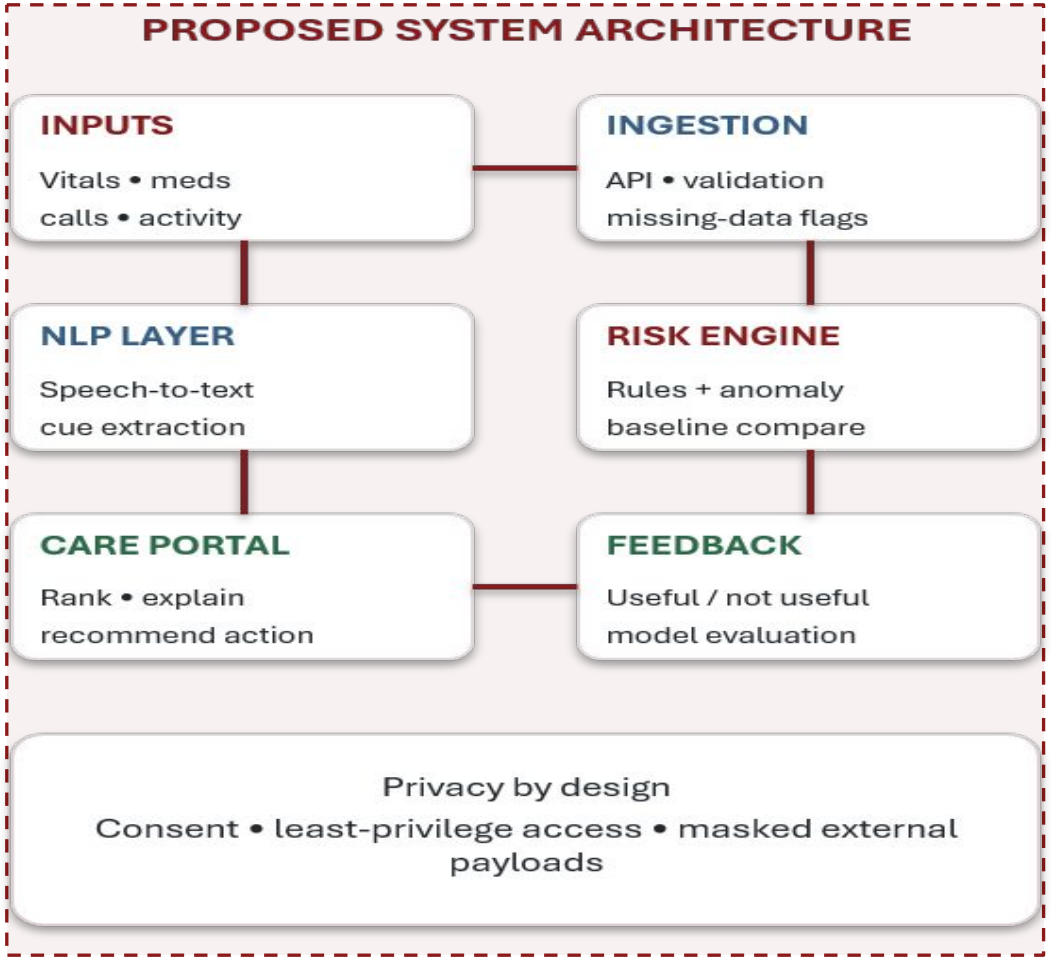
TECHNICAL APPROACH

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Technology Stack and Methodology

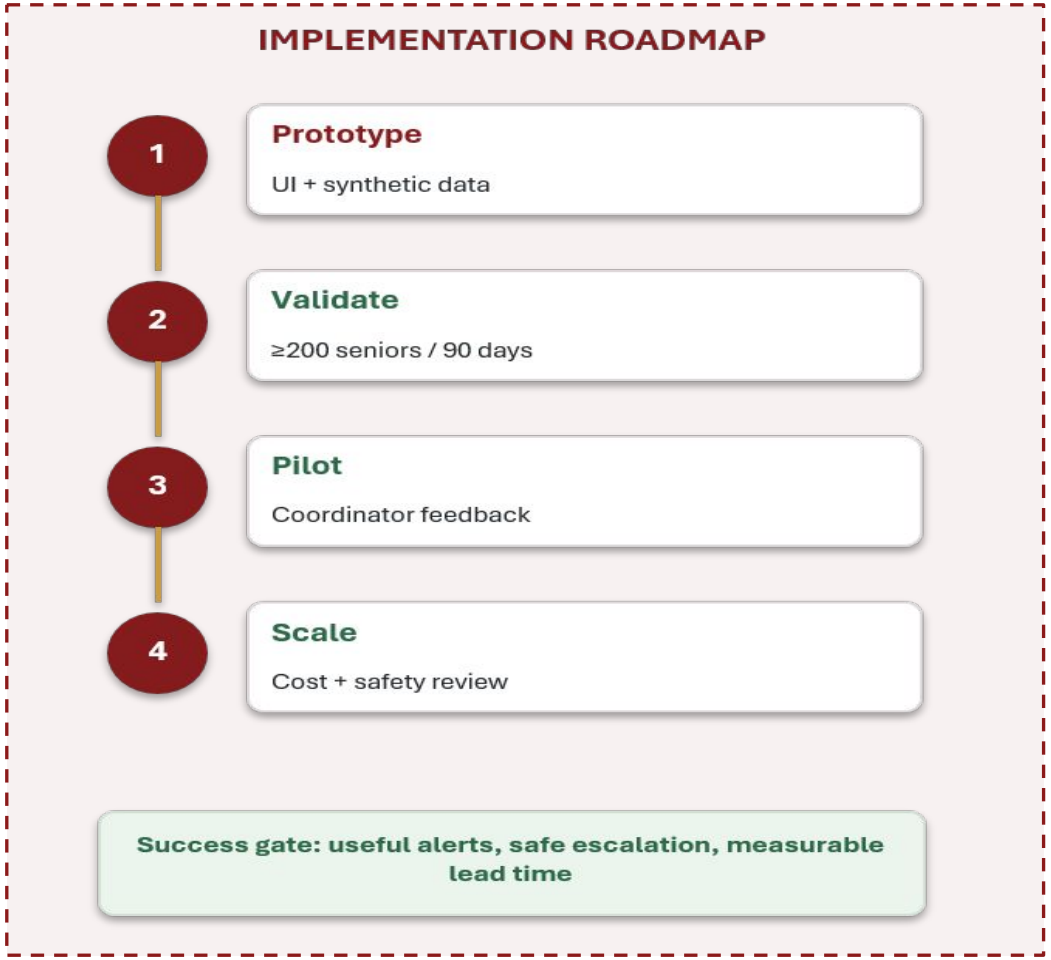
- **Full-Stack Architecture:** React / responsive web senior UI, Kotlin-ready mobile flows, and FastAPI Python microservices with role-based auth.
- **Multi-Tier Risk Engine:** Rule-based safety floors + statistical anomaly detection (Z-score vs 14-day rolling baseline) + lightweight ML scoring.
- **Bilingual NLP & Voice:** Speech-to-text pipeline with prompt-engineered cue extraction detecting pain, dizziness, appetite loss, and confusion.
- **Resilient Fallback Chain:** Cloud primary → on-device/cached rules → automated PSTN voice check-in when senior device goes offline.
- **Robust Synthetic Cohort:** Simulated 200+ seniors over 90 days with realistic missingness, multi-morbidity decline, and acute fall edge cases.
- **Explainability & Human Loop:** Every alert includes transparent contributing factors; coordinator feedback fine-tunes ongoing threshold models.



FEASIBILITY AND VIABILITY

Can It Be Built and Sustained?

- **Demonstrated Working MVP:** Functional end-to-end loop: live senior check-in, NLP cue extraction, dynamic risk triage, and coordinator escalation.
- **Frugal Cost at Scale:** Optimized scheduled batch scoring and localized cached inference achieve estimated cost < ₹25 ($0.30) / senior / month.
- **Eliminating Alert Fatigue:** Personal baselines, confidence scoring, and multi-factor confirmation reduce false alarms to 0.89 / week.
- **Low-Tech & Offline Access:** Automated interactive voice calls (IVR/PSTN) and large one-tap buttons guarantee access without digital literacy.
- **Privacy & Data Safety:** DPDP Act compliant: role-based access, end-to-end data encryption, granular family consent, and audit logging.
- **Clear Phased Deployment:** Rapid transition from 200-senior synthetic benchmark → 50-senior supervised pilot → regional coordinator scaling.



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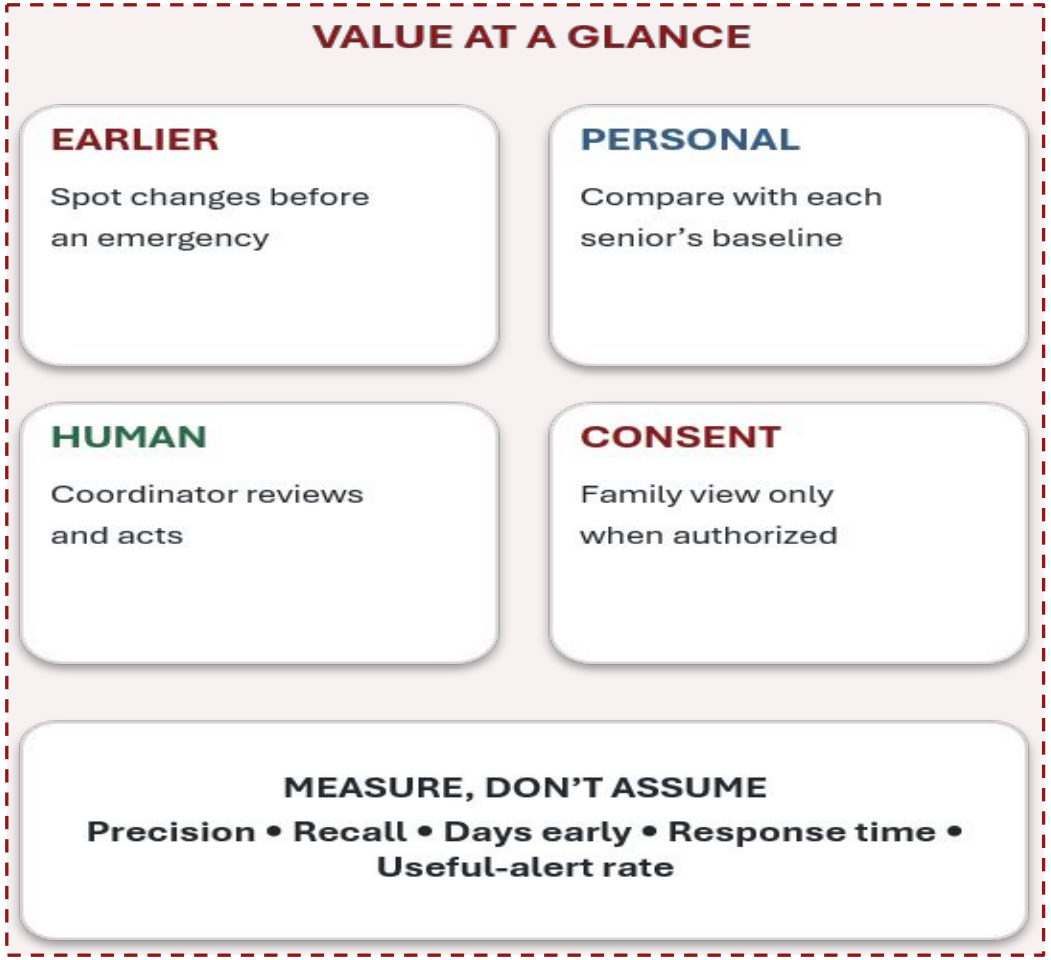
IMPACT AND BENEFITS

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Value Created

- **100% Critical Recall:** 100% of planted acute decline and emergency fall scenarios caught with 91.5% overall alert precision.
- **9 Days Median Warning:** Detects multi-day health and behavioral deterioration median 9 days before acute hospitalization.
- **Low False-Alarm Burden:** Achieves 0.89 false alarms / senior / week, protecting coordinators from notification fatigue.
- **Coordinator Superpowers:** Prioritized triaging queue with transparent clinical reasoning saves 45+ minutes of daily manual review.
- **Peace of Mind for Families:** Dignified, consent-governed updates keep remote children informed without intrusive surveillance.
- **Human-in-the-Loop Safety:** Assists care coordinators with timely alerts without ever generating automated medical diagnoses.



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RESEARCH AND REFERENCES

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Research Work and Sources

Problem grounding

- **Samarth Life Management Brief:** Designed strictly around elderly constraints (low digital literacy, polypharmacy, mobility risks) and care coordinator workflow requirements.

Proposed technical references

- **PhysioNet Clinical Datasets:** Physiological baselines & vital sign trend modeling (MIMIC/WFDB benchmark sources).
- **MobiFall & SisFall Datasets:** Public accelerometer and gyro movement patterns used for fall impact vs ADL calibration.
- **AI4Bharat & IndicSTT:** Open Indian-language speech models for Hindi & Hinglish voice check-in parsing.

DATA & EVALUATION PLAN

SYNTHETIC COHORT BENCHMARK

200 seniors × 90 days • Realistic missingness & sensor noise • Planted gradual cognitive, vital & mobility declines.

MODEL & ARCHITECTURE DISCLOSURE

Transparent deterministic floor rules + rolling 14-day anomaly engine • Full audit logs and parameter documentation.

RIGOROUS EVALUATION METRICS

100% Critical Recall • 91.5% Precision • 9 Days Median Early Warning • 0.89 False Alarms/Wk • Coordinator usefulness rating